



Nov 3 — Write down your first and last name

Econ 139: Labor Economics

Technology: A Task Based Approach

David Arnold



A Look Back

- In 19th century England 1 out of 4 workers were employed in the textile industry
- During this time period, we began to see the first forms of mechanization in the form of mechanical weavers
 - According to legend, Ned Ludd, a skilled weaver, became enraged at the new technology and smashed it in a fit of fury
- A political movement formed: The Luddites

Technology and labor markets

- For a long time, there have been two key arguments on how technology will impact workers
- **The Optimist View** – what matters is productivity. If technology increases productivity, this will be realized as wage gains for workers

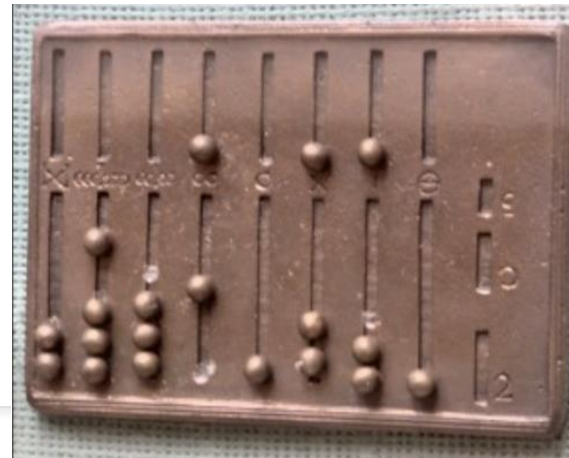
Technology and labor markets

- For a long time, there have been two key arguments on how technology will impact workers
- **The Optimist View** – what matters is productivity. If technology increases productivity, this will be realized as wage gains for workers
- **The Pessimist View** – Technology may increase productivity, but the gains are shared unequally. You could have the richest become richer while the displaced workers are made worse off

A Brief History of Computing

- There are many places we could start when discussing the impact of technology on the labor market
 - Industrialization, electrification, etc.
- We will focus on **computers**. This technology has had arguably the largest impact through the latter half of the 20th century into the present
- So let's begin the discussion with what a computer actually does

Computers



- Computers are at their core information processing machines
 - Definition: an information processing machine is one that receives data, processes it using some set of rules, and then outputs some information
- One of the earliest information processing machines was the abacus, which has been around since antiquity for the purpose of collecting taxes

Charles Babbage

- One of the revolutions in computing was the invention of the programmable computer by Charles Babbage
 - Prior to this machines were more or less hardwired to perform a single task
 - Babbage realized that if you also include a way to give instructions, multiple tasks could be performed
 - This is the birth of programming



THE LATE MR. BABBAGE.

Alan Turing

- While Babbage's computers were programmable, they were quite simple, being able to perform a few operations, like addition, subtraction, multiplication
- Alan Turing's work was mathematical and conceptual: he asked, can you create a machine that can simulate any computation another machine can perform? (the concept of a machine has a mathematical meaning in his framework)
- The answer was yes. This was an astounding finding, with enormous real-world consequences



Universal Turing Machine

- We take for granted computers exist and have tremendous capabilities. But it was not at all obvious that such a machine was even theoretically possible
- To see this, let's take some early information processing systems

Universal Turing Machine

- Census tabulating machine. Sorts and counts data



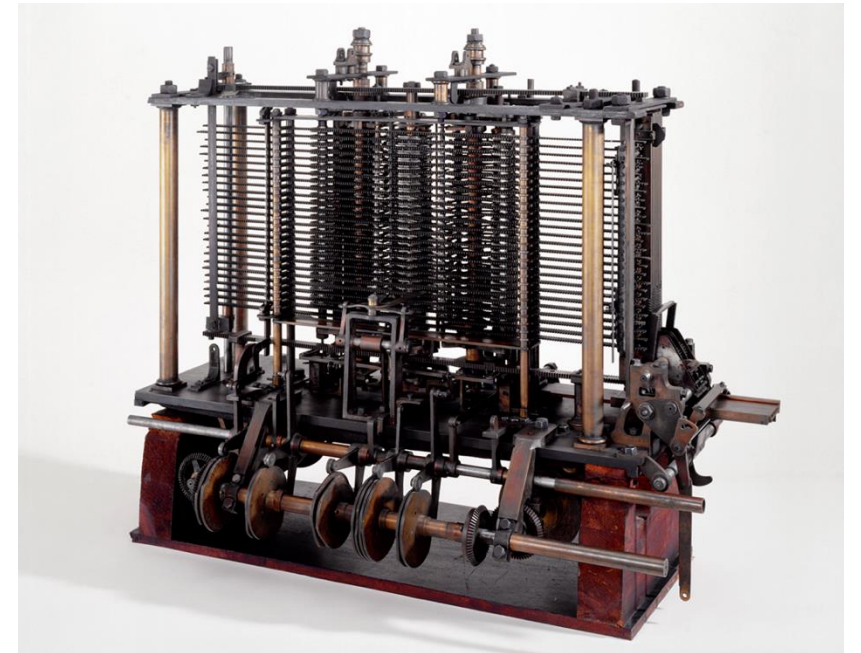
Census tabulating machine

Universal Turing Machine

- Census tabulating machine. Sorts and counts data
- Babbage's analytical machine. Could perform addition, subtraction, multiplication



Census tabulating machine



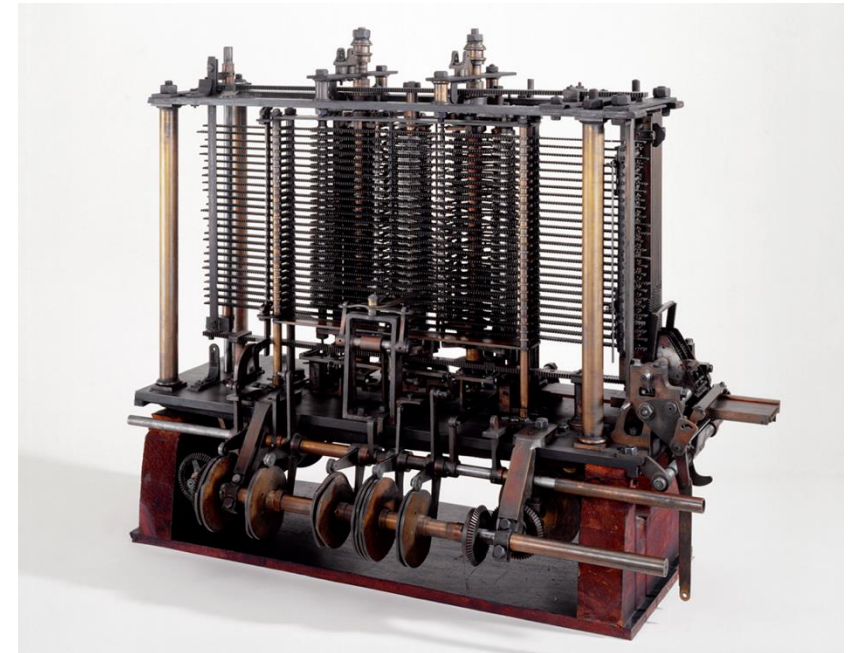
Babbage's Analytical Engine

Universal Turing Machine

- Not at all obvious that you can build one machine that does both
- Turing's insight implied you can build a very general machine that can simulate the behavior of both these machines (and any machine)



Census tabulating machine



Babbage's Analytical Engine

A timeline



A Case Study

- Imagine you are an accountant in 1980s
- A task for the day could be the following: calculate payroll
 - For each worker, add up how many hours they work and multiply by their wage and add to payroll ledger
 - Simple task, but can take an extremely long time
 - Probably need to double check – errors can be costly

A Case Study

- Imagine you are an accountant in 1980s
- A task for the day could be the following: calculate payroll
 - For each worker, add up how many hours they work and multiply by their wage and add to payroll ledger
 - Simple task, but can take an extremely long time
 - Probably need to double check – errors can be costly
- Now imagine your firm gets Excel. Now calculating payroll takes a few keystrokes

Two Potential Reactions

- As an accountant you could have two potential reactions
 - This is amazing! A tedious task that used to take hours can be done in minutes
 - This is terrifying. If the computer can do this quickly, maybe the firm doesn't need so many accountants anymore...

Two Potential Reactions

- As an accountant you could have two potential reactions
 - This is amazing! A tedious task that used to take hours can be done in minutes
 - This is terrifying. If the computer can do this quickly, maybe the firm doesn't need so many accountants anymore...
- This probably sounds a bit familiar regarding today's discussions around AI
 - This is certainly not new. If you told a weaver in 19th century England that a tiny fraction of employment would be in textiles in 100 years, they might imagine something catastrophic has happened

So What Actually Happened in the Past?

- So theoretically, computers could be good or bad for accountants
- We are going to solidify this intuition with a conceptual framework: the task-based model of production
- Then we will look into the data and see what actually happened, not just to accountants, but to all types of workers

The Task Model

- The task model comes from Autor, Levy, and Murnane (2003) and has become an extremely influential way to think about technology impacts on the labor market
- Prior to this model, most models thought about firms using different types of labor as input: high-skilled, medium-skilled, low-skilled
- The task model is going to change a bit how we think about a production function. Workers produce tasks, tasks are combined to output. This intermediate step is going to lead to some interesting insights and guide our empirical analysis

Routine vs. Nonroutine Tasks

- Before we get to actual tasks, it will be helpful to classify tasks by a few characteristics
- Routine tasks: tasks that can be accomplished by a set of clearly defined rules

Routine vs. Nonroutine Tasks

- Before we get to actual tasks, it will be helpful to classify tasks by a few characteristics
- Routine tasks: tasks that can be accomplished by a set of clearly defined rules
- Nonroutine tasks: tasks that require some flexibility and adaptability

Cognitive vs. Manual

- Before we get to actual tasks, it will be helpful to classify tasks by a few characteristics
- Cognitive tasks: tasks that require some thought

Cognitive vs. Manual

- Before we get to actual tasks, it will be helpful to classify tasks by a few characteristics
- Cognitive tasks: tasks that require some thought
- Manual tasks: tasks that require physical labor

Types of Tasks

	Routine	Nonroutine
Cognitive	Recordkeeping, performing calculations	
Manual		

Types of Tasks

	Routine	Nonroutine
Cognitive	Recordkeeping, performing calculations	
Manual	Sorting items, assembling items, some production tasks	

Types of Tasks

	Routine	Nonroutine
Cognitive	Recordkeeping, performing calculations	Medical diagnoses, legal writing, managing a team
Manual	Sorting items, assembling items, some production tasks	

Types of Tasks

	Routine	Nonroutine
Cognitive	Recordkeeping, performing calculations	Medical diagnoses, legal writing, managing a team
Manual	Sorting items, assembling items, some production tasks	Cleaning a house, making food, driving a truck

Types of Tasks

	Routine	Nonroutine
Cognitive	Recordkeeping, performing calculations	Medical diagnoses, legal writing, managing a team
Manual	Sorting items, assembling items, some production tasks	Cleaning a house, making food, driving a truck

- What do we think computers can do?
- (Until recently) computers rely on having clearly programmable set of rules to follow, so excelled in routine tasks
- Machine Learning Revolution may be changing this by being able to perform some nonroutine cognitive tasks
- Today we focus on traditional personal computers

Data Sources

- Autor, Levy and Murnane (2003) use the Dictionary of Occupational Titles
 - This was a massive effort U.S. Department of Labor to classify occupations.
 - It was later replaced by O*NET (which we will cover in a future data lecture)

DOT for Accountant

Work activities in this group primarily involve the application of the principles of accounting, cost analysis, and statistical analysis to problems of fiscal management, auditing, and the like. Typically, workers are engaged in devising accounting systems and procedures; appraising assets and evaluating costing methods, investment programs, and monetary risks and rates; and preparing statistical tabulations and diagrams and financial reports, statements, and schedules for use by management.

- DOT Accountant (1965)

Impact of Computers on Accountants

- Let's pick out one part from the description
preparing statistical tabulations and diagrams and financial reports, statements, and schedules for use by management.
- To perform this task, need some inputs (i.e. the raw data that goes into statistical tabulations). Therefore, to accomplish this is a mix of routine and nonroutine tasks
 - Routine: Gathering data, inputting data
 - Nonroutine: Choosing analysis, performing analysis, and preparing reports

Impact of Computers on Accountants

- There are a couple different forces at work here:
 - **Replacement:** some parts of the tasks (i.e. statistical computations), can be performed by computers. Maybe you need fewer accountants now
 - **Productivity:** the accountant will be more productive. Can produce more reports, more accurate reports, quicker. Maybe accountants are so productive now their role is expanded into things like forecasting, budgeting, etc.

Impact of Computers on Support Staff

- For support staff (for example data entry), unclear what tasks remain
- Before computers, keeping organized (paper) spreadsheets, inputting new data, was a tremendous task
- Now this can be done for computers, so demand for support staff may fall

Empirics

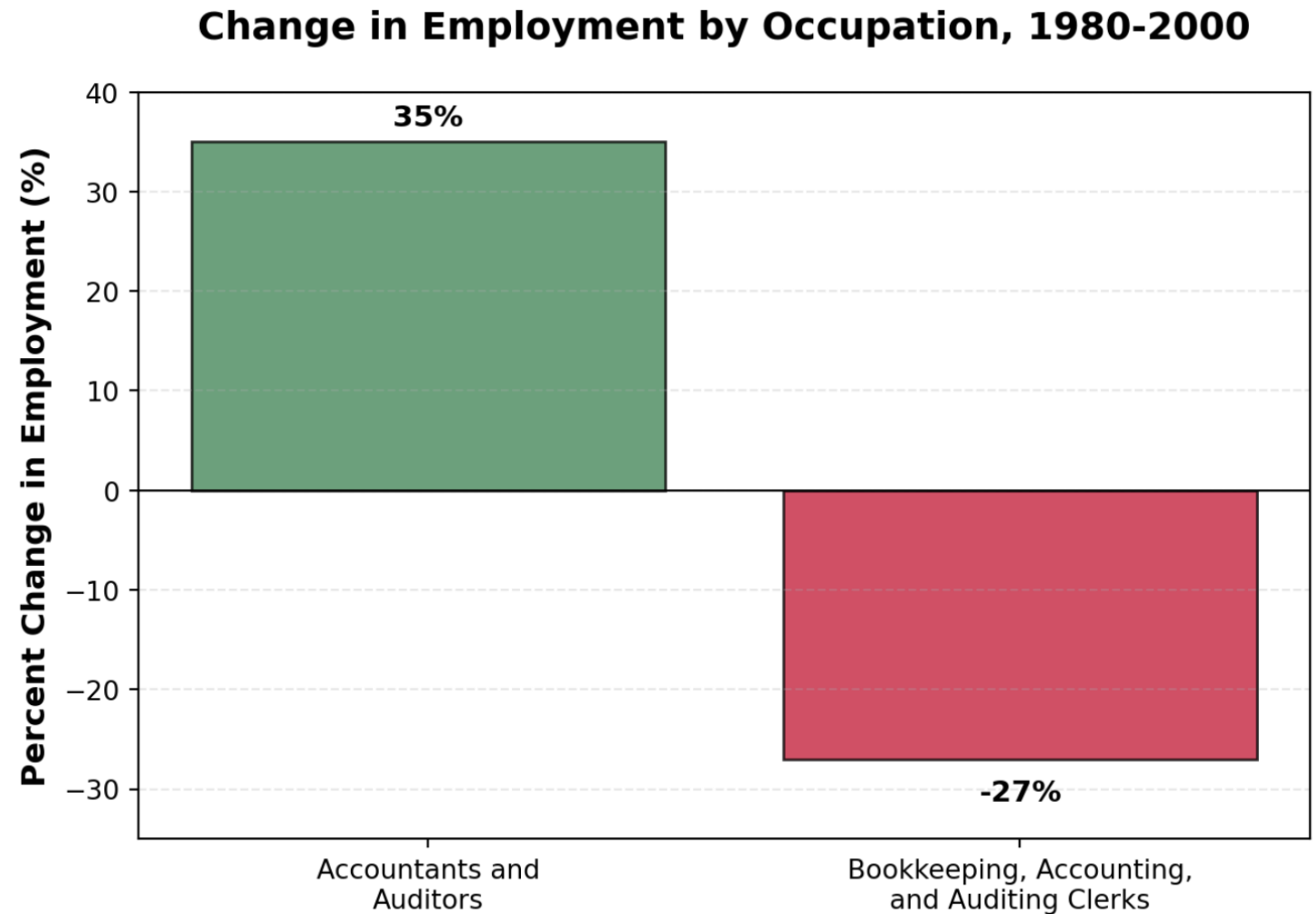
- For the impact of technology on workers, it is often easy to tell different stories
 - You can tell a story in which tasks are replaced and main impact is decline in demand for workers
 - You can also tell a story in which productivity gains are so big that even if some tasks are replaced, overall demand for workers goes up
- At the end of the day this is an empirical question, so let's see what actually happened in the data. **Did demand for accountants grow or decline due to the introduction of computers**

Census

- We are going to again use data from the Census. This again comes from the IPUMS USA database
 - Our sample will be using the 2000 Census and comparing to the 1980 Census
- This data has a 5% random sample of the US workforce
- Includes information on occupation, so we can compute the change in the number of accountants between 2000 and 1980 (as well as the change in support staff)

Census data

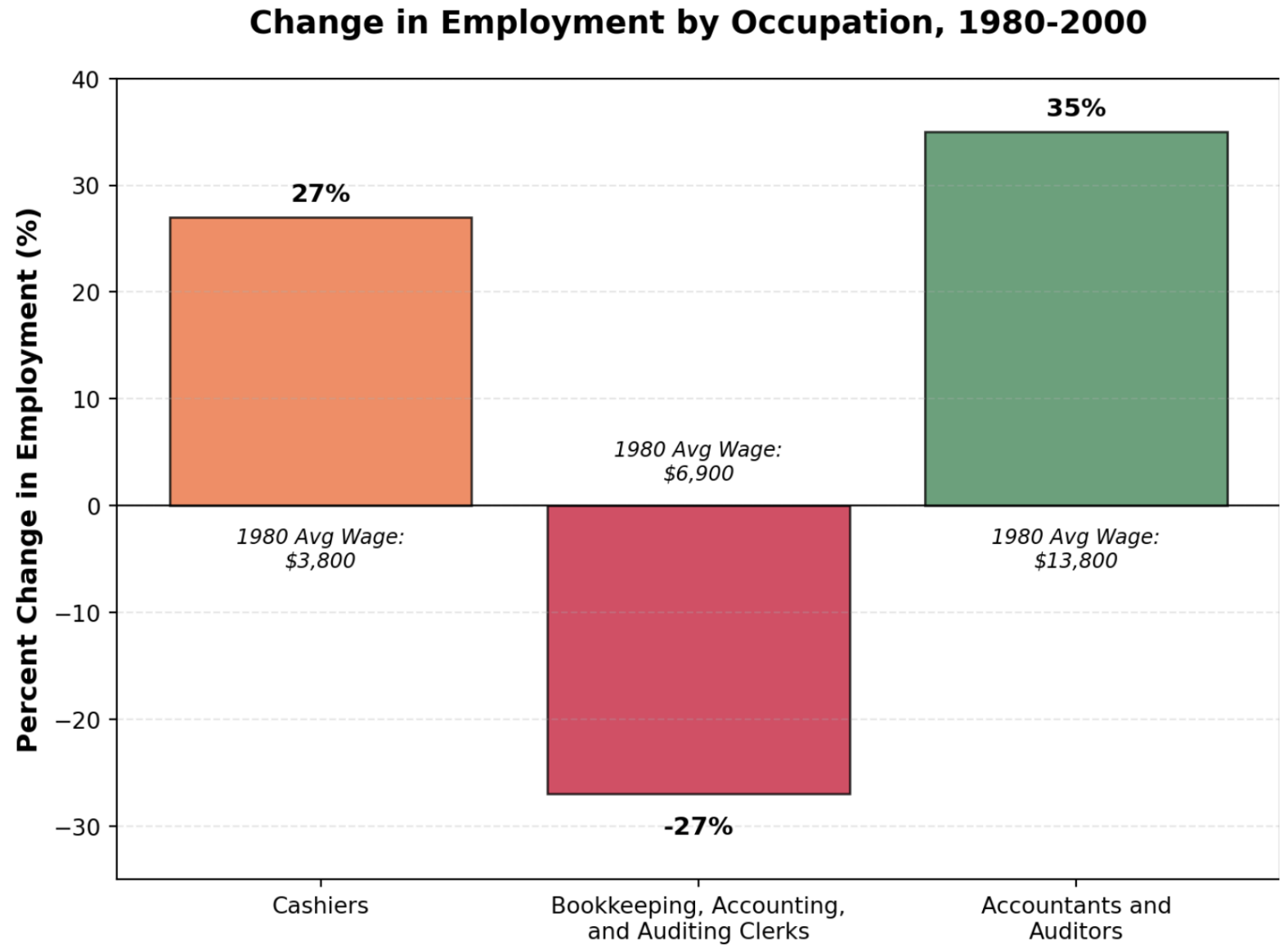
- Accountants and auditors saw large increases in employment
- Support staff (bookkeeping, accounting and auditing clerks) saw large decreases



Job Polarization

- Where do the people who would have become support staff (or who got laid off) go?
- Well, how about a profession that is not exposed to computers?
- For example, in-person checking out of customers not really able to be automated at this time. So let's see what is going on with cashiers generally during this period

Job Polarization



Tasks

- Let's map this back to our task-based framework

Occupation	Primary Task Type	Can Primary Task be done by Computers?
Accountants	Cognitive Nonroutine	No

These are just summaries/predictions. Most occupations have a mix of cognitive, manual, routine and nonroutine

Tasks

- Let's map this back to our task-based framework

Occupation	Primary Task Type	Can Primary Task be done by Computers?
Accountants	Cognitive Nonroutine	No
Cashiers	Manual Nonroutine	No

These are just summaries/predictions. Most occupations have a mix of cognitive, manual, routine and nonroutine

Tasks

- Let's map this back to our task-based framework

Occupation	Primary Task Type	Can Primary Task be done by Computers?
Accountants	Cognitive Nonroutine	No
Cashiers	Manual Nonroutine	No
Accounting Clerks	Cognitive Routine	Yes

These are just summaries/predictions. Most occupations have a mix of cognitive, manual, routine and nonroutine

Tasks

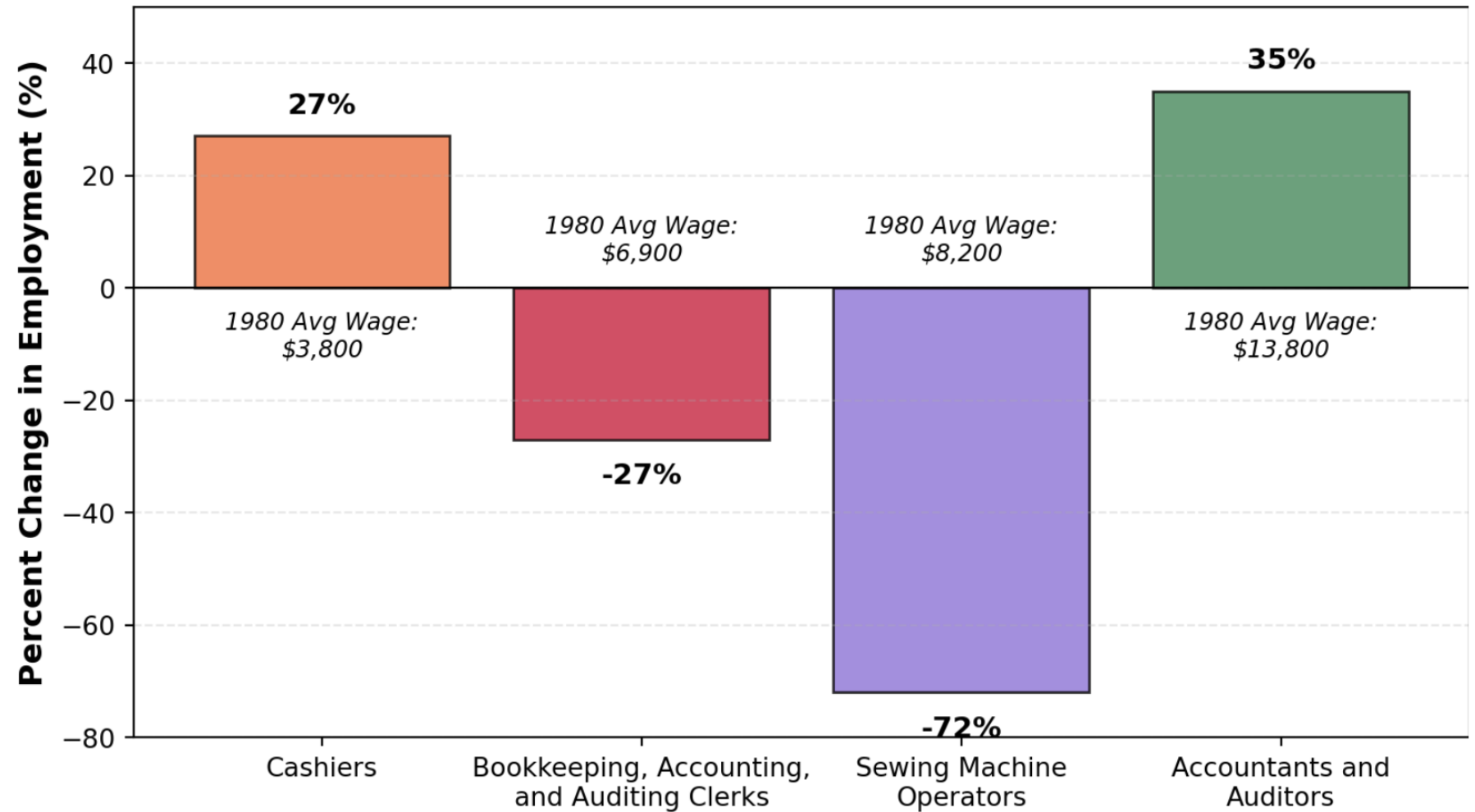
- Let's map this back to our task-based framework

Occupation	Primary Task Type	Can Primary Task be done by Computers?
Accountants	Cognitive Nonroutine	No
Cashiers	Manual Nonroutine	No
Accounting Clerks	Cognitive Routine	Yes
Sewing Machine Operators	Manual Routine	Yes

These are just summaries/predictions. Most occupations have a mix of cognitive, manual, routine and nonroutine

Job Polarization

Change in Employment by Occupation, 1980-2000

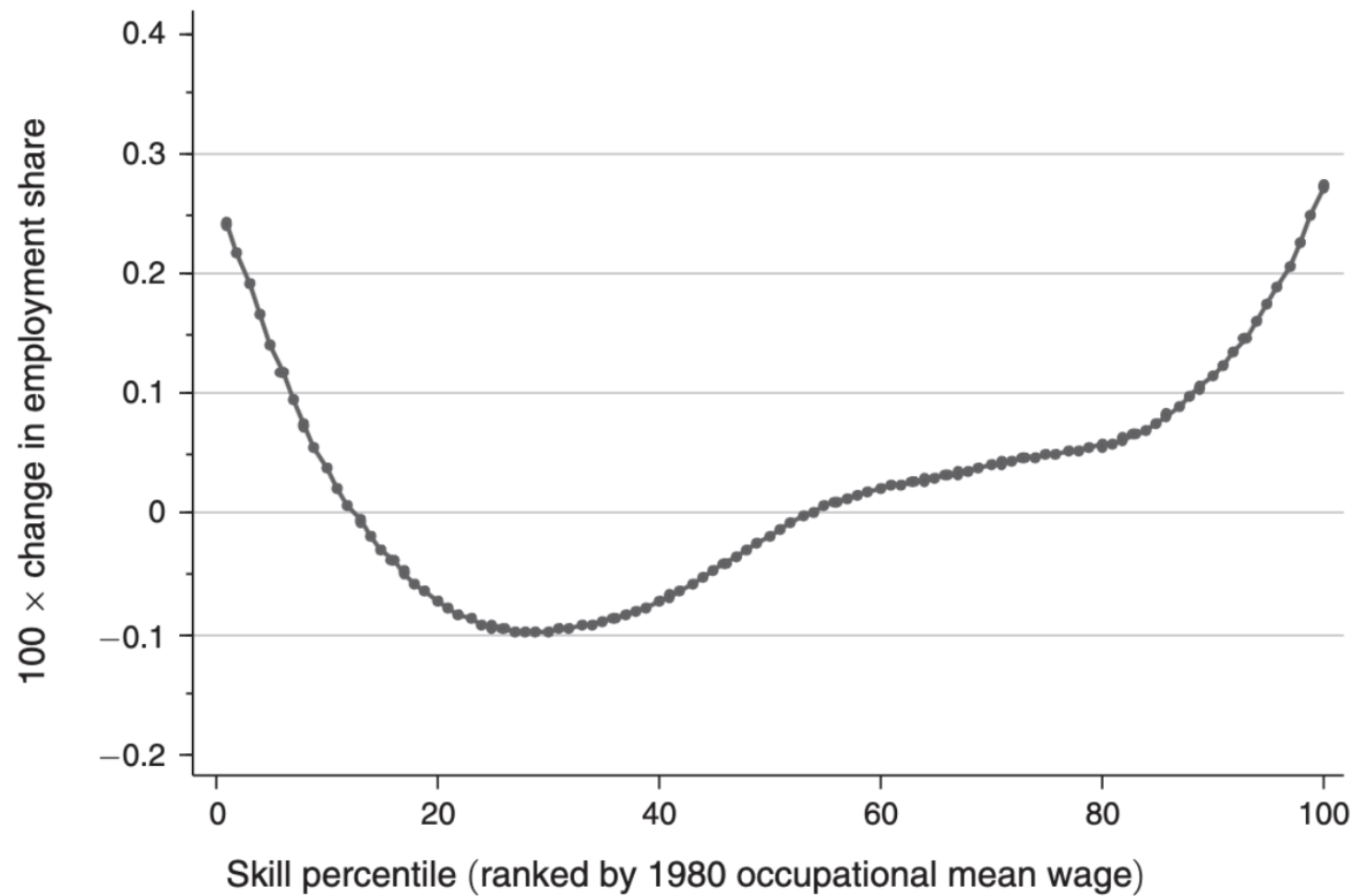


Generalizing

- Summary so far: we found middle-income jobs which tend to be routine saw large declines in employment
- Consistent with computers causing job polarization which leads to widening inequality
- But we have done this for only four occupations. We need to make sure this is a more general pattern

Generalizing

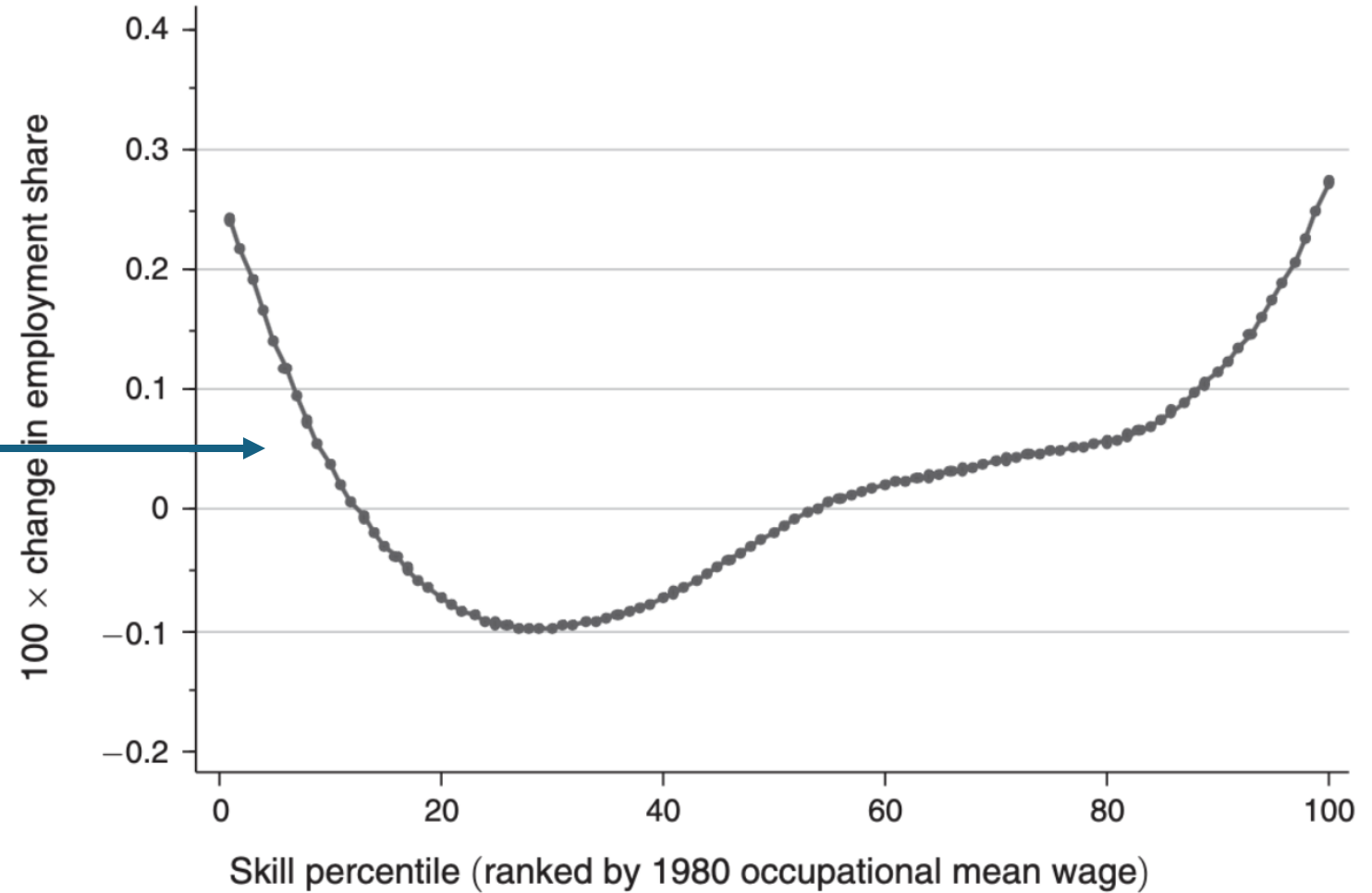
Panel A. Smoothed changes in employment by skill percentile, 1980–2005



Generalizing

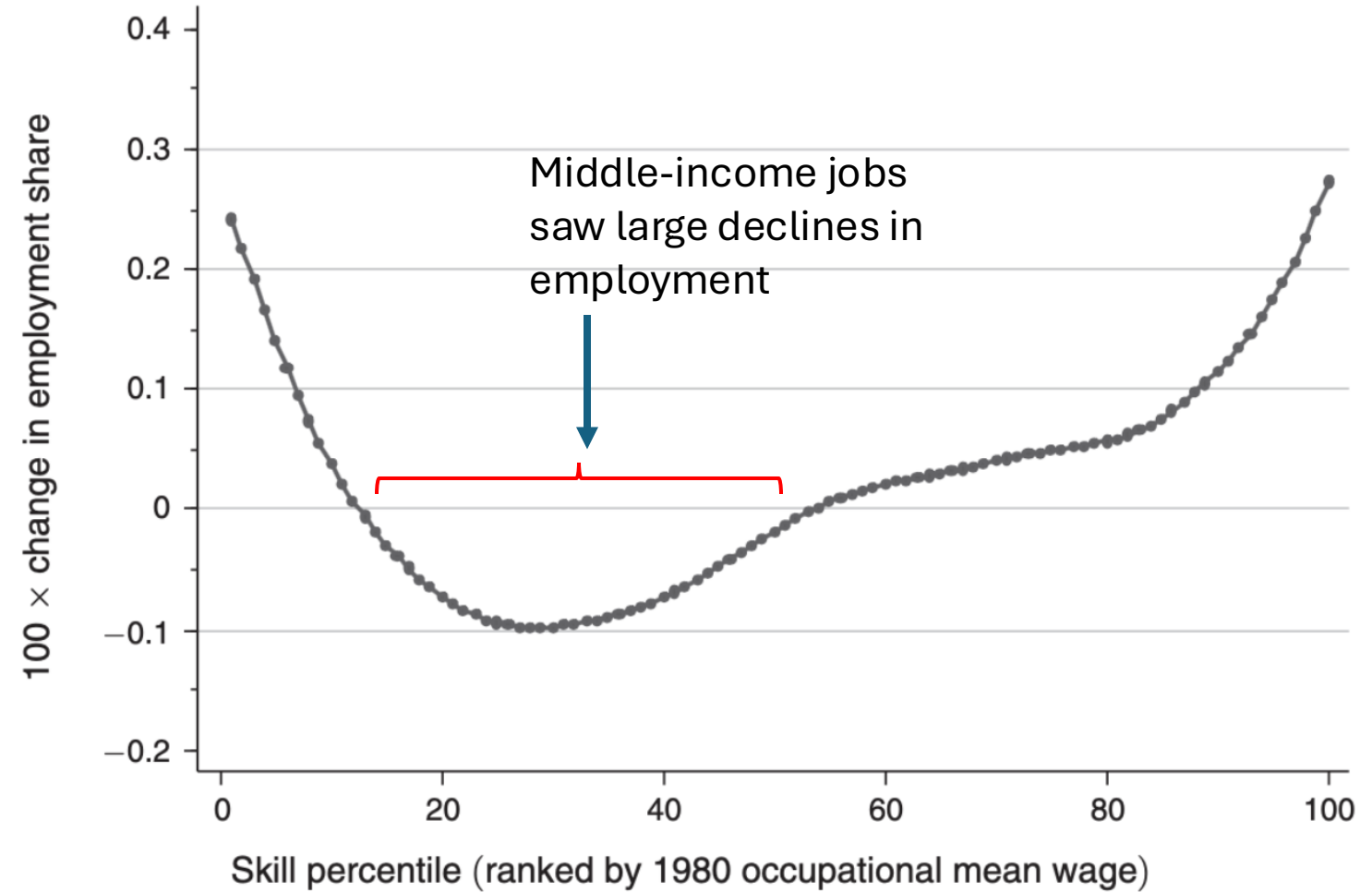
Low wage jobs saw large increases in employment

Panel A. Smoothed changes in employment by skill percentile, 1980–2005



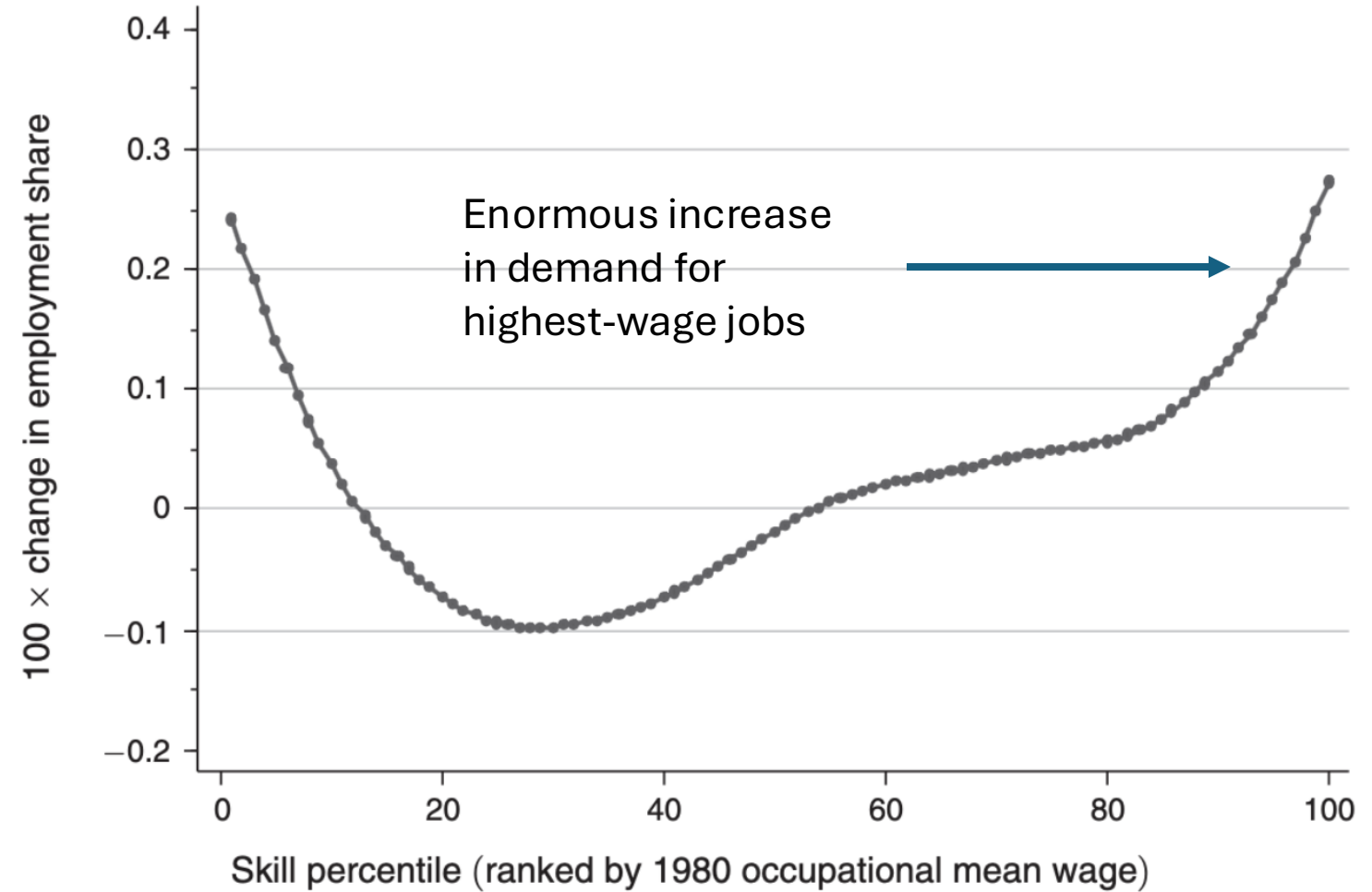
Generalizing

Panel A. Smoothed changes in employment by skill percentile, 1980–2005

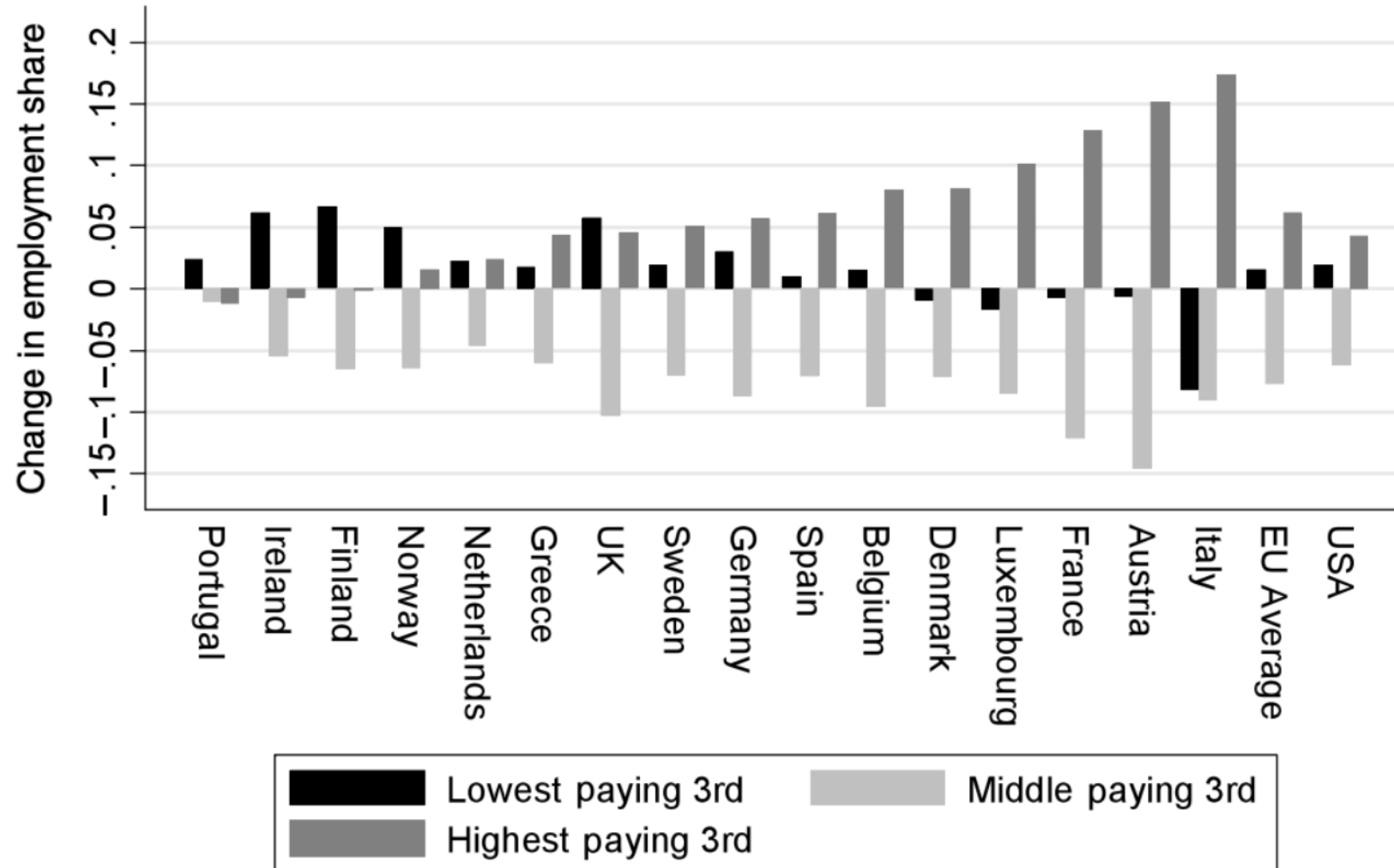


Generalizing

Panel A. Smoothed changes in employment by skill percentile, 1980–2005



Change in employment shares by occupation 1993-2006 in 16 European countries
Occupations grouped by wage tercile: Low, Middle, High



Same pattern
consistent across
many countries

Back to Tasks

- Let's once again link these changes back to tasks
- Acemoglu and Autor (2011) consider 4 groups
 - Professional, Managerial and Technical
 - Clerical, Sales and Administrative Support
 - Service Occupations
 - Production, Craft, Repair and Operative

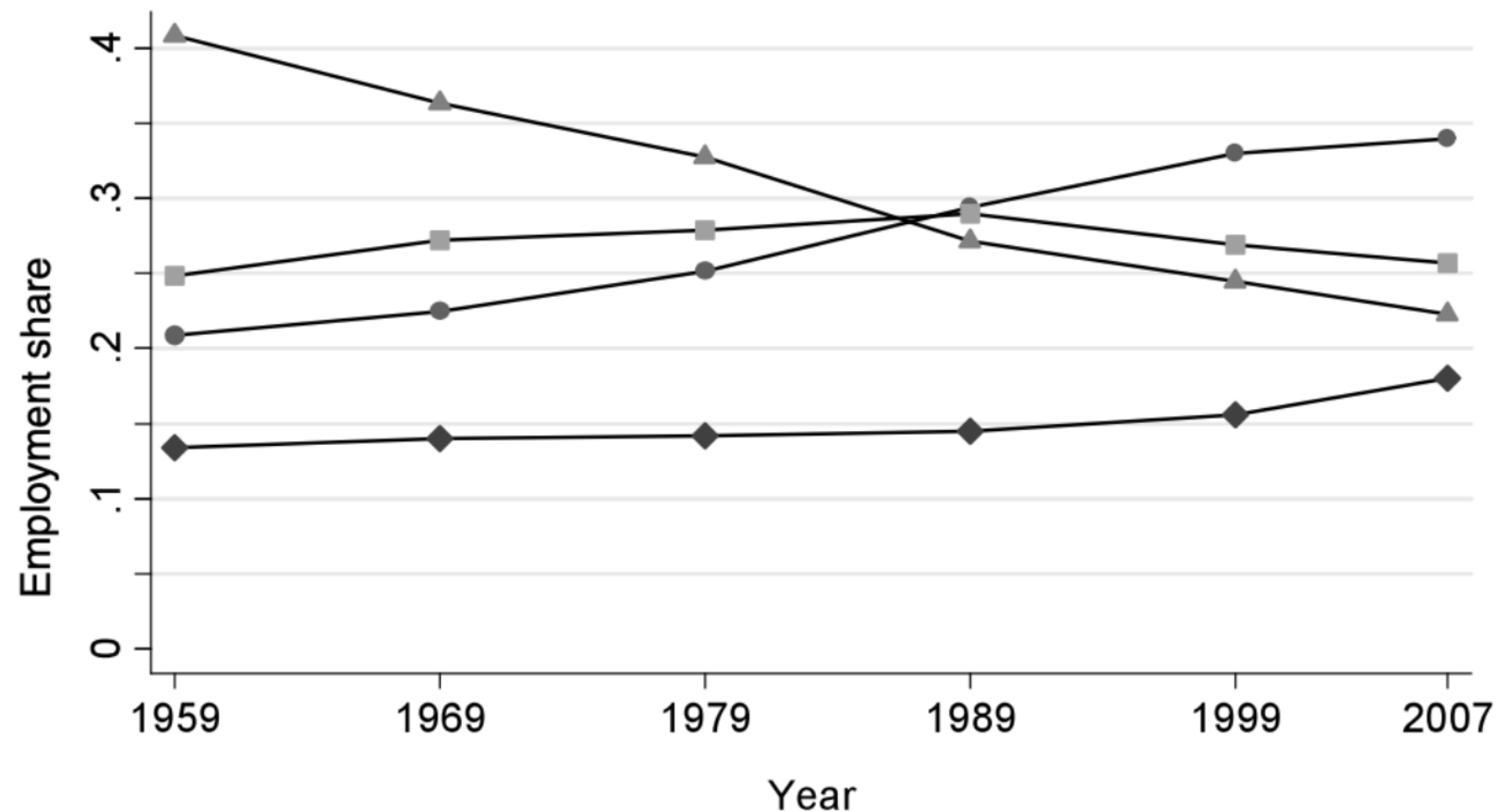
Table 7.1: Task Classification and Predicted Employment Changes

Group Label	Example Occupations	Routine/Non-Routine	Cognitive/Manual	Predicted Change
Professional, Managerial, and Technical	Managers, Engineers, Lawyers, Doctors	Non-Routine	Cognitive	↑ <i>Increase</i>
Clerical, Sales and Administrative Support	Bookkeepers, Secretaries, Data Entry	Routine	Cognitive	↓ <i>Decrease</i>
Service Occupations	Food Service, Janitors, Personal Care	Non-Routine	Manual	↑ <i>Increase</i>
Production, Craft, Repair and Operative	Assemblers, Machine Operators, Laborers	Routine	Manual	↓ <i>Decrease</i>

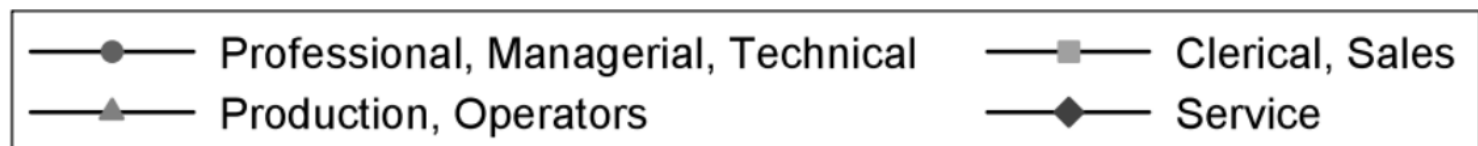
Employment shares by major occupation groups, 1959-2007:



Males and females



(a)



Summary

- Task-based framework provides a powerful lens to understand changes in employment during the computer age
- Next we will consider how much technology has contributed to the big trends we started the class with
 - Technology has contributed to all three
- Also consider some (speculative) thoughts about the future impact of AI on the labor market